

Tut 9

$\vec{u}$

Ex. 11.3

Q29 If $\underline{u}_1$ and $\underline{u}_2$ are ^①orthogonal ^②unit vectors
and $\underline{v} = \textcircled{3} a\underline{u}_1 + b\underline{u}_2$,

find $\underline{v} \cdot \underline{u}_1$.

✓ ① $\underline{u}_1$ and $\underline{u}_2$ are orthogonal $\Leftrightarrow \boxed{\underline{u}_1 \cdot \underline{u}_2 = 0}$

② $|\underline{u}_1| = 1$ and $|\underline{u}_2| = 1$

✓ ③ $\underline{v} = a\underline{u}_1 + b\underline{u}_2$

$$(\underline{x} + \underline{y}) \cdot \underline{w}$$

$$= \underline{x} \cdot \underline{w} + \underline{y} \cdot \underline{w}$$

$$\underline{x} = a\underline{u}_1, \underline{y} = b\underline{u}_2$$

$$\underline{v} \cdot \underline{u}_1 = (a\underline{u}_1 + b\underline{u}_2) \cdot \underline{u}_1 \quad \text{by } \textcircled{3}$$

$$= (a\underline{u}_1) \cdot \underline{u}_1 + (b\underline{u}_2) \cdot \underline{u}_1$$

$$= a(\underline{u}_1 \cdot \underline{u}_1) + b(\underline{u}_2 \cdot \underline{u}_1)$$

$$= a|\underline{u}_1|^2 + b \cdot 0 \quad \because \underline{u}_1 \cdot \underline{u}_1 = |\underline{u}_1|^2$$

$$= a|\underline{u}_1|^2$$

$$= a \cdot 1^2 \quad \text{by } \textcircled{2}$$

$$= a$$

and $\left(\begin{array}{l} \underline{u}_1 \text{ and } \underline{u}_2 \\ \text{are orthogonal} \end{array} \right)$
or (by $\textcircled{1}$)